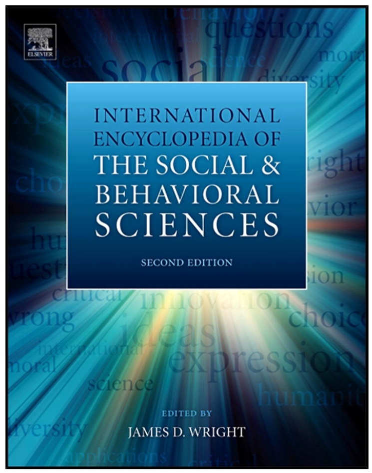

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Stress in Adolescence: Effects on Development

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Abstract

Adolescents frequently have to cope with various stressors that could be potential threats to their healthy development and well-being. Most important mental health consequences of stress in adolescence, such as depression, anxiety, suicide, substance use, and antisocial behavior are reviewed. Physiological mechanisms through which stress exerts its effects on health, as well as some physical health outcomes, such as somatic symptoms, immune changes, and illnesses (cancer, type 1 diabetes, and dermatological conditions) are described. Furthermore, the effects of cognitive appraisal, coping, and social support as mediating and moderating factors between stress and various health outcomes also are explained.

Adolescence is an important period of human development characterized by neurobiological, hormonal, psychological, and social changes. During this developmental period, adolescents have to cope with various common stressors, including physical and sexual changes related to puberty, school life demands, problems with initiating and maintaining friendships and romantic relationships, career choice and beginnings of their working life, and a gradual independence from their families. Additionally, some teenagers have to cope with more unusual stressors that could be related to family (e.g., mental or physical illness, death of a loved one, drug or alcohol abuse, parental separation or divorce), social problems (e.g., poverty, violence), as well as their individual problems (e.g., teenage pregnancy, serious illness, school failure, and various kinds of abuse). Therefore, adolescence often has been thought of as a time of storm and stress, a period of rebellion characterized by conflicts with parents. Contrary to the traditional considerations, however, more recent research findings support the view that storm and stress is not a universal and exclusively biologically based phenomenon (Susman and Rogol, 2004). Contemporary understanding suggests that adolescence is a period when specific types of stressors are more likely to occur than in other developmental periods and that these developmental changes also include adaptation to cultural expectations of becoming an adult. Nonetheless, adolescents frequently have to cope with various stressors that could be potential threats to their healthy development and well-being. For example, cumulative and chronic stressors may lead to increased emotional distress (e.g., depression, anxiety), antisocial behaviors (e.g., aggression, delinquency), risky health behaviors (e.g., substance use, risky driving), and physical health outcomes (e.g., physical symptoms, common colds). Traumatic stressors, such as physical, sexual, and relational victimization, may result in posttraumatic stress symptoms. Furthermore, many studies have found that stressful experiences exert short-term as well as long-term effects on adolescents' healthy development.

Mental Health Consequences of Stress in Adolescence

Depression

Research on recent birth cohorts showed that adolescence is one of the developmental periods during which depression most frequently occurs (Hammen, 2009). Although estimates of childhood depressive episodes for children under the age of 12 years vary from 1 to 3%, several studies found adolescence lifetime prevalence of major depression to be from 9.3 to 24.0% (Merikangas and Knight, 2009), with an additional 11% of adolescents reporting minor depression (Hammen and Rudolph, 2003). Longitudinal studies have found that depression in adolescence might have long-term negative consequences regarding some other mental disorders (e.g., anxiety disorders, substance use) and various negative outcomes (e.g., academic failure, teenage pregnancy, conflicts with family and peers, and suicide). Additionally, it has been found that among those with early onset of depression, very high percentages experience depressive episodes as adults and that early onset of depression is related to a wide range of psychiatric, physical, and functional health outcomes (Abela and Hankin, 2008).

Although the majority of stressful events do not lead to depressive episodes, most depressive episodes are preceded by stressful events. Exposure to stressful events, such as parental death, serious illness, and sexual and physical abuse, as well as less traumatic events and hassles are frequently found to be risk factors for depressive symptoms and disorders in both clinical and community samples of children and adolescents. For example, a review of 60 studies exploring the associations between stressors and psychological problems, including depressive symptoms in children and adolescence showed that stressful experiences predict increases in mental health problems over time (Grant et al., 2004). It has been found that stress and depression are bidirectionally influenced, that is, those with depression histories have greater chance to experience additional distress in the future. A better understanding of the reciprocal influence of stress and depression may

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have important implications for the course, treatment, and prevention of depression.

Strong evidence exists for gender differences in depressive disorders in adolescence. Recent meta-analysis reveals a higher prevalence of depression among teenage adolescent girls compared with boys (Costello et al., 2006). One longitudinal birth-cohort study found gender difference in major depressive episodes, with girls having higher rates of depression beginning from age 13 years, with this difference becoming more pronounced by age 15 years (Hankin et al., 1998). Therefore, it seems that the gender difference in depression emerges between ages 12 and 15 years, and by early adulthood women are twice as likely as men to experience depressive episodes. Several biological and psychosocial factors that can explain this difference have been identified. Besides biological risk factors (e.g., dysregulation of the hypothalamic-pituitary-adrenocortical (HPA) axis, genetic factors, and dysregulation of neurotransmitter systems regulating mood), psychosocial risk factors (e.g., negative cognitive style, interpersonal orientation, and negative life stress) also proved to be important (Hilt and Nolen-Hoeksema, 2009). Namely, many studies demonstrated that girls are exposed to greater levels of stressors, as well as that they report more interpersonal stressors compared with boys who report more academic stressors (Hammen, 2009). Girls also more often than boys are the victims of physical and sexual abuse, with both types of abuse being risk factors for the development of depression. Additionally, girls appear to generate more dependent stressors that contribute to their depression, and they are more reactive to stress than boys. Research results consistently show that under high levels of stress, girls have greater tendency to express depressive symptoms than boys.

Anxiety Disorders

Anxiety and depression share important features, but are nevertheless distinguishable phenomena. Although most fears during childhood and adolescence are considered appropriate for the developmental periods and the situations in which they occur, and although anxiety usually is thought of as adaptive because it helps in anticipating threat or harm, its clinical expression usually refers to developmentally inappropriate fears or to those that are developmentally appropriate but that produce excessive distress or dysfunction (Moore et al., 2010). Anxiety disorders usually begin relatively early in life, often manifesting before late adolescence.

Epidemiological data show that the anxiety disorders often are evident in children and adolescents. Overall, lifetime prevalence estimates in children and adolescents for any anxiety disorder range from approximately 10 to 20% (Costello et al., 2004). Specific types of anxiety disorders, such as panic disorder (PD), specific phobias, social phobia, obsessive-compulsive disorder (OCD), posttraumatic stress disorder (PTSD), acute stress disorder, separation anxiety disorder (SAD), and generalized anxiety disorder (GAD) occur with different frequency during the period of childhood and adolescence. Epidemiological studies show that GAD is the most common, followed by SAD, social phobia, and the specific phobias. Much less frequent are PTSD, OCD, and PD, with less than 1% prevalence in population samples. Considering gender differences, research shows that girls are affected more, and the overall gender ratio for anxiety disorders throughout childhood and adolescence is ~2:1 (Costello et al., 2004). Untreated anxiety disorders tend to become chronic, with only a minority of adults (12–30%) exhibiting spontaneous remission. Adolescents show somewhat higher rates of recovery, and the majority of early onset anxiety disorders remit by early adulthood.

Early anxiety disorders, however, especially those that develop in late adolescence, represent major risk factors for the development of anxiety disorders in adulthood. Several disorders, such as PD, social phobia, GAD, and OCD, frequently manifest waxing and waning symptoms in which the exacerbation of symptoms may be related to stressful experience (DSM-IVTR; APA, 2000).

Ample evidence indicates that the most prominent factors that contribute to the development and maintenance of anxiety symptoms and disorders are biological predispositions (e.g., family-genetic, neurobiological, and hormonal), but environmental factors should not be underestimated. Namely, although more consistent findings were obtained for depressive disorders, many studies showed that children and adolescents who are exposed to chronic stressors and unpredictable circumstances may be at greater risk for developing anxiety disorders. Research suggest that GAD may result after family relocations, school changes, or other stressful life events (Vasey and Ollendick, 2000), although stressful events related to health and family discord are predictive of increases in anxiety sensitivity (McLaughlin and Hatzenbuehler, 2009). Parenting practices also may lead to anxiety disorders. For example, overprotective parents usually reinforce avoidant and inhibited behavior in their children encouraging them to perceive many situations as new or negative, which in turn lead to their impaired ability to cope and raises their anxiety. Furthermore, a moderate degree of specificity exists in the associations between types of life events and types of anxiety disorders. For example, it has been found that parental separation was related with GAD, while parental death was related with increased risk for phobia (Kendler et al., 1992).

Prevailing theoretical models of the causes of internalizing disorders (depression and anxiety) are variants of diathesis-stress models. The fundamental assumption of a diathesis-stress model is that both the diathesis (also interchangeably referred to as vulnerability) and various environmental conditions (e.g., stressors) are necessary for the development of a disorder and that neither one alone is sufficient. Therefore, it is assumed that internalizing symptoms results from the interaction between different vulnerabilities (e.g., neurobiological, genetic, cognitive and environmental) and stressful events.

Suicide

According to the World Health Organization (2001), suicide is one of the three leading causes of death among people from 15 to 34 years of age in all countries. The results of a meta-analysis of 128 studies from different countries showed that 10% of youths reported a suicide attempt, and 20–30% reported suicidal ideations at some point in their life (Evans et al., 2005). The higher rate of death by suicide is observed in adolescent boys, while after puberty adolescent girl show higher rates of suicidal ideation and suicide attempts.

The strongest risk factors for attempted and completed suicide in youth are previous suicide attempts, suicidal ideation, and the presence of psychiatric disorders, especially depressive disorder, alcohol and drug abuse, conduct disorder, and aggressive-impulsive behavior (Gould et al., 2003).

Furthermore, environmental stressors also have been recognized as having a role in suicide attempts and suicidal ideation. For example, several studies have found that life events, such as interpersonal loss and legal or disciplinary problems have been linked to completed suicide (Jacobson and Gould, 2009). Childhood physical abuse (e.g., corporal punishment), sexual abuse, and neglect also have been related to suicidal behavior. The younger the person was at the time of the first abuse, the higher the number of suicide attempts reported (Berk et al., 2010).

Bullying also is linked to suicidal behavior among adolescents, especially girls, as indicated by a large epidemiological study showing that bullying perpetrators and their victims, as well as youth who are both victims and perpetrators, have a higher risk for suicidal ideation and suicide attempts (Brunstein-Klomek et al., 2007). Parental conflict is found to be related more frequently to suicide in younger adolescents, whereas romantic relationship difficulties are associated more strongly with suicide in older adolescents (Brent et al., 1999).

Neither psychiatric disorders nor stress alone, although important, are sufficient to trigger suicidal behavior. It is likely that the combination of underlying psychiatric disorders paired with an acute stressor typically leads to suicide.

Substance Use

Adolescence is a period of life characterized by the experimentation with risky behaviors and is a critical developmental period for the onset of substance use and, sometimes, the beginning of the trajectory toward substance dependency. Evidence shows that approximately 50% of adolescents have tried an illicit drug at least once before the end of their high school education, more than 75% have tried alcohol, and approximately 60% of high school seniors report some experience with cigarette smoking, with 19% of them being daily smokers (Johnston et al., 2001, 2002). Substance use and addictive disorders are topics of great importance because of their pervasive negative consequences that may affect the course of adolescent development as well as outcomes in adulthood. For example, accidents, suicides, and homicides account for more than 80% of adolescence deaths, of which at least 50% include drugs and alcohol (Soderstrom and Dearing-Stuck, 1993). Studies also suggest that long-term consequence that is most reliably related to substance use in adolescence is the continued use of that substance. It has been found that adolescence substance use is associated with psychological distress and mental health problems in adulthood.

Research shows many risk factors for drug abuse, each having a different impact depending on the phase of development. These factors can be related to individual characteristics (e.g., genetic susceptibility to addiction, high sensation seeking and impulsivity, alienation, academic failure, and low commitment to school), as well as environmental stressful events (e.g., family problems, problems with peers and community environment).

Studies consistently have found that adolescents who experience high levels of life stress are more likely to use alcohol or drugs and to increase the quantity and the frequency of substance use over time (Chassin et al., 2004). The most frequent explanation of the role of stress in substance use is that adolescents who experience high level of stress and negative affect use alcohol or drugs as a way to decrease them. This is in accord with research findings showing that emotional distress, negative affect, and clinically diagnosed depression are associated with adolescent substance use, although some results show that negative affect might be a result, rather than a cause, of adolescent substance use.

Furthermore, although substance use often is considered as a motivation to reduce negative affect, it also is seen as a way to increase positive affect (Chassin et al., 2004). This distinction in the motive for substance use is important because it has been suggested that adolescents who use substances to maintain or enhance positive affect may use them moderately in quantity and frequency, whereas those who use substances to decrease negative affect may use them more excessively.

Antisocial Behavior

Among several types of mutually intercorrelated antisocial behaviors the most important are conduct disorder, aggression, and delinquency. They are related logically and empirically and, therefore, risk factors that apply to one of them also likely apply to the other two (Farrington, 2004). The prevalence estimates for antisocial behaviors in adolescence are usually very high. For example, the prevalence of conduct disorders in adolescent boys varies from 6 to 16%, and in adolescent girls it ranges from 2 to 9% (Mandel, 1997).

The etiology of antisocial behaviors includes multiple risk factors that do not occur in isolation. They can be grouped into individual, family, and extrafamilial factors (Connor, 2002). Genetic factors and individual differences in temperament are the most important individual factors. Among family and extrafamilial risk factors, chronic and acute stressful life events are predominant. Several family factors found to be strongly associated with antisocial behaviors are severe marital discord, overcrowding or large family size, paternal criminality, maternal psychiatric disorder and the family history of admission of a child into the care of local child protective services. Extrafamilial risk factors for antisocial behaviors include relationship with peers, social deprivation, neighborhood violence, and media violence (Connor, 2002). Except for heritable genetic factors, the majority of risk factors are nonspecific. Namely, they increase the risk for wide variety of general adolescent psychopathology, including different types of antisocial behaviors. Research also suggest that cumulative effects of multiple risk factors exert a much stronger influence on antisocial behavior in children and adolescents than any specific type of risk factors.

Physiological and Physical Health-Related Consequences of Adolescent Stress

When an event has been appraised as stressful, our body triggers a sequence of physiological reactions. Physiological stress response and its control are regulated by the two axes of the neuroendocrine system: HPA and sympathetic-adrenal-medullary (SAM) pathways. The activation of the SAM axis is related strongly to acute emotions, such as fear and anger, and describes the first part of the stress response. During stress, the hypothalamus helps the body to prepare for 'fight or flight' by triggering sympathetic impulses to various organs, stimulating the release of epinephrine, norepinephrine, and other catecholamines in the blood circulation. Catecholamines can cause changes in blood pressure, heart rate, sweating, and pupil dilatation, which result in a feeling of arousal. They also have an effect on a variety of body tissues and can lead to changes in immune function. In addition to the sympathetic activation, stress also triggers changes in the HPA system. At the same time that sympathetic activity increases, the hypothalamus releases corticotropin-releasing hormone, which stimulates the anterior pituitary gland to secrete adrenocorticotropic hormone (ACTH), which increases the adrenal cortex's secretion of cortisol. Cortisol supplies cells with amino acids and extra energy sources and diverts glucose from skeletal muscles to brain tissue. Stress also stimulates the release of glucagon from the pancreas, growth hormone from the anterior pituitary, antidiuretic hormone from the posterior pituitary gland, and renin from the kidneys. Through feedback to the hypothalamus, epinephrine can increase secretions of ACTH to higher levels and thus can maintain the increased activation of HPA axis (Tsigos and Chrousos, 2002). The neurohormonal and immune changes that develop during stress are physiological pathways through which stress can lead to mental and physical health outcomes.

Many factors during adolescence simultaneously contribute to the heightened sensitivity to stressful events, especially neurobiological processes, such as the continued maturation of stress-responsive brain regions, the changes in hormonal reactivity, and the quantity and quality of stressful events. For example, research shows that gender differences in the functioning of stress-related physiological pathways begin during adolescence and persist into adulthood. Available evidence suggests that HPA response in girls differs from those in boys, generally, and is more under the influence of genetic factors and previous environmental exposures. Studies suggest that variants in specific genes, parental depression, and stressful events affect cortisol responses in girls more than those in boys. A greater reactivity to internal and external influences under stress than may be adaptive in many situations also might be related to risks (e.g., increased probability of depression) (Oldehinkel and Bouma, 2011).

Most studies dealing with the impact of stress on adolescent health and psychological well-being have found that adolescents suffer from a large number of body complaints that might be considered as psychosomatic symptoms (e.g., headaches, stomachaches, indigestion, circulatory problems, and nausea). Among those most frequently reported are abdominal pain, chest pain, and headaches (Seiffge-Krenke, 1998). There are indications that daily hassles are better predictors of somatic symptoms than major life events. Among stressful situations that are most important for elevated somatic symptoms in this developmental period are school-related stressors, such as interpersonal problems with peers and teachers, schoolwork pressure, and fear of school failure. Additionally, the transition from primary to secondary school, as well as conflicts with parents, are stressful experiences that in many adolescents increase the frequency and intensity of somatic health symptoms.

Stress also leads to a significant decrease in the function of cellular immunity such as lower NK-cells activity, lower lymphocyte proliferation, and production of gamma interferon (IFN-γ), as well as the increases in herpes virus antibody production. These changes are related to a greater susceptibility to infectious diseases, such as herpes reactivations, infections of the upper respiratory tract, and HIV infections. Exam stress, a stressor typical for adolescents, frequently leads to these immune changes and related infectious diseases (Kiecolt-Glaser et al., 2002).

Research also shows that stressful life events have an impact on chronic illnesses in adolescents. Especially high incidence of stressful life events has been found in children and adolescents with cancer, who most frequently reported the experience of marital separation, death of a close family member, change in residence, and change in health of family members before the onset of cancer (Seiffge-Krenke, 1998). Although these findings could not be considered as evidence of causal relations between stressful events and diseases, and although physiological mechanisms of these associations are not fully understood, it seems that not a single event, but rather the accumulation of several major stressors, could have negative effects on health outcomes.

Although the strongest associations between life events and the onset of chronic illness have been found in cancer patients, the negative impact of life events also was found in patients with other chronic diseases. Studies on diabetic patients also show greater frequency of stressful life events before the onset of diabetes, and it has been found that stress could contribute to the onset of type 1 diabetes through its effects on the immune function (Cox and Gonder-Frederick, 1992). Furthermore, higher levels of diabetes-specific stress as well as general stress are associated with less adherent behavior and subsequently more problems with metabolic control, at least in patients with type 1 diabetes (Farrell et al., 2004).

Stress can be an important factor in the onset, clinical course, and asthma-related mortality. For example, stressful life events such as family dysfunction and patterns of behavior that increase psychological conflict as well as stressful school exams can accelerate the onset and recurrence of asthma after the remission. Furthermore, studies have found that children and adolescents from socially isolated families have more frequent symptoms of asthma, more days of restricted activities because of the symptoms, worse illness management, and more frequent visits made to physicians (Wright et al., 1998). It seems that high levels of ongoing chronic stress plus an acute negative life event could be an especially important risk combination for an asthma attack in children and adolescents. Research highlights immune mechanisms through which psychosocial variables, such as stress, may plausibly affect asthma outcomes. For example, stress has been linked to inflamed bronchial pathways implicated in asthma (Chen et al., 2006).

Psychosocial stress also has been related to the onset or worsening of various dermatological disorders. It may influence dermatological symptoms in primary stress-reactive skin disorders in which neuroendocrine and psychoneuroimmunologic factors have a crucial role (e.g., psoriasis, atopic dermatitis or eczema, urticaria, acne, and alopecia areata), in disorders characterized by physiological response (e.g., excessive perspiration and blushing), as well as in skin disorders resulting from psychiatric condition such as OCD, PTSD, or major depressive disorder (e.g., dermatitis artefacta, neurotic excoriations, or trichotillomania) (Gupta, 2010).

Stressful events frequently experienced by adolescents, such as exams, have been found to worsen some dermatological diseases (e.g., acne). Additionally, an increase in distress is related with the increase in acne severity. Acne has a peak incidence during adolescence, a period typical for intensive preoccupation with body image and appearance. Therefore, besides the contribution of psychosocial stress to the onset of dermatological disease, the disease itself may result in significant stress for the patient. For example, in vulnerable adolescents, even mild acne could result in severe depression.

Several neurophysiological mechanisms have been implicated in stress-mediated skin disorders. Because stressful emotional experiences disturb the regulation of the HPA and the SAM systems, they may result in the alternations of glucocorticoid levels. For example, studies have found a relation between stress and an imbalance in the epidermal permeability barrier function, which is mediated by the increase in endogenous glucocorticoids. Additionally, higher anxiety in acne patients have been associated with high blood catecholamine levels, which decreased with the treatment of the acne, suggesting that the anxiety in acne patients is associated with the SAM system activation (Gupta, 2010).

Effects of Cognitive Appraisal, Coping, and Social Support on the Experience of Stress

The same stressful event in different individuals may lead to the distress of various intensities. What is extremely stressful in one individual may not be at all stressful in another. Among the most important sources of variation in the effects of stress during adolescence are personality traits, cognitive appraisal, coping with stress, and social support. According to the transactional model of stress and coping, the appraisal of a situation determines the degree of stress experienced by the individual (Lazarus and Folkman, 1984). The model describes two main types of cognitive evaluations of stressful situations, primary and secondary appraisals. Primary appraisal includes the individual's assessment of the stressfulness of the situation. Situations appraised as stressful can involve harm, loss, threat, or challenge. Secondary appraisal includes the individual's assessment of controllability of the stressful episode and the choice of coping strategies. These two types of appraisals are parallel processes that may be automatic and usually occur without conscious awareness. Coping is most often defined as constantly changing cognitive and behavioral efforts to manage specific external or internal demands that are appraised as taxing or exceeding the resources of the person (Lazarus and Folkman, 1984). According to this theory, coping efforts are broadly classified into two types: problem-focused and emotion-focused coping. Problem-focused coping can be directed outward to alter some aspect of the environment or inward to alter some aspect of the self. They include cognitions and behaviors directed toward analyzing and solving a problem and may as well include dividing the problem, seeking information, and considering alternative courses of actions as well as direct action. The efforts directed toward self are various forms of reappraisals, such as changing the meaning of the situation or event, reducing ego involvement, or recognizing the existence of personal resources or strengths. Emotion-focused coping strategies are directed toward decreasing emotional distress. They include various ways of coping usually described as expression and suppression of emotions as well as seeking emotional support from others. Besides problem- and emotion-focused coping, avoidance coping also is included in some taxonomies of coping with stress. It refers to cognitive or behavioral avoidance or denial of the problem.

During puberty and adolescence, the differentiation of coping resources, skills, and strategies oriented toward problem solving are under the influence of cognitive development and social experience. Adolescence is a period in human life characterized by a relatively quick development and the use of a broader range of coping strategies, as well as the tendency to abandon ineffective coping. For example, between 13 and 15 years of age, a significant decrease of avoidance coping has been noted. Also, it has been observed that older adolescents use wishful thinking and distancing less than before. Adolescence is also a critical period for the development of some ineffective coping strategies, especially social withdrawal and alcohol and substance use (Aldwin, 2007). Regarding gender differences, the most consistent result found in the literature is that girls use emotion-focused coping more often than boys. To the lesser extent, this is also true for problem-focused coping. This usually is explained by their faster maturation and the experience of higher levels of stress during early adolescence when compared with boys. Also, it has been found that boys use drugs, alcohol, and repressive coping more often than girls.

Among major antecedents of coping with stress are individual differences in temperament and other personality traits as well as in self-concept. These factors influence the availability of coping strategies in stressful situations as well as the appraisal of the stressfulness of events. For example, the majority of research shows that the personality trait of extraversion is related to active and more efficacious coping strategies, such as problem-focused coping, whereas neuroticism is related to passive and less efficacious coping strategies, such as avoidance. Neuroticism is related to higher exposure and reactivity to stressful events and also is considered to be a disposition for stressful experience even in the absence of stressors. Furthermore, adolescents who are high in self-esteem usually adopt problem-focused coping strategies, whereas adolescents with low self-esteem often adopt avoidant and emotion-focused coping strategies.

Coping strategies are not defined as generally effective or ineffective, but rather their effectiveness may depend on the type of situation, specific combinations between different coping strategies, and timing in their use during the course of a stressful transaction. For example, avoidance strategies are more effective in reducing emotional distress in the short term, and problem-focused strategies are more effective over the long term. When the stress is perceived as controllable, problem-focused coping strategies are associated with fewer psychological symptoms, whereas in uncontrollable stressful situations, emotion-focused coping is related to fewer symptoms.

Considering various health consequences of coping with stress, numerous studies show that some coping strategies more often are related to positive, whereas others to negative, health outcomes. For example, longitudinal research on adolescents reveals that the use of avoidance and emotion-focused coping have been linked with more intensive depressive and anxiety symptoms, whereas the use of problem-focused coping has been associated negatively with depression and hopelessness. Furthermore, problem-focused coping has been found to correlate negatively with health problems and health risk behaviors, whereas avoidance coping correlated positively with these domains (Wilson et al., 2005). Generally, it could be concluded that when adolescents lack adequate coping skills, there is a risk from various negative consequences, including psychological distress and suicidal attempts, smoking and substance use, and high-risk sexual behaviors (Rew, 2005).

Social support is considered to be one of the most important psychosocial factors that influence the relationship between stress and health outcomes. It has been defined broadly as interpersonal social resources that involve either the presence or the implication of stable human relationships. Social support is regarded as structural (e.g., number of social contacts, size, and density of social networks) or functional (e.g., informative, emotional, instrumental). Evidence suggests that social support generally contributes to better adjustment of adolescents and that their low satisfaction with social support is associated with anxiety, depression, psychosomatic symptoms, and interpersonal sensitivity.

The protective roles of social support on health outcome are explained by the two models. The main-effect model states that social support has independent effects on mental and physical health outcomes regardless of the level of stress experienced, while the stress-buffering model posits that social support has indirect effects on health outcomes by acting as a buffer against stress (Cohen and Wills, 1985).

Among adolescents, the social support of family and peers has great significance. Parental support has been found to be a predictor of better adjustment and less distress in adolescence. The presence of familial social support has been associated with higher levels of resilience. On the other hand, different studies show that two characteristics of a risky family social environment – conflict and aggression and a cold, unsupportive, or neglectful home – have adverse developmental effects on adolescents. It has been shown that parental support, involvement, and monitoring reduce the probability that adolescents will engage in a variety of health-compromising behaviors. Furthermore, longitudinal studies have found that lack of support and deficient nurturing during childhood leads to higher rates of illness and physical complaints several years later (Repetti et al., 2002).

Without the support of a peer group, adolescents experience loneliness and a sense of lack of belonging and acceptance, which, in turn, can contribute to frustration and feelings of rejection and emergence of antisocial behavior. Along with positive effects of the peer support, adolescence is also a period during which relationships with peers can undermine healthy development. For example, drug use and risky sexual behavior may be influenced greatly by peers. Regarding drug use, adolescents who become involved with substance use usually have friends who do the same.

Social support most effectively acts as a buffer against stress when it matches the needs elicited by a stressful event. Even social relationships that are close and supportive are not uniformly positive and may have negative consequences. For example, being close to and having highly supportive antisocial parents might result in various types of problematic behaviors of adolescents, such as substance abuse or conduct disorder (Rook, 2003). Also, it has been found that diabetic adolescents who turn to their peers for coping strategies may have worse metabolic control than those who turn to their parents (Aldwin, 2007).

Cross-Cultural Differences in the Effects of Stress on the Development of Adolescents

As mentioned, developmental changes during adolescence, among other factors, also include adaptation to cultural expectations of becoming an adult. The greatest part of our knowledge about the effects of stress on development in adolescence was drawn from the research on the samples from Western societies. Theoretical assumptions as well as empirical evidence suggest the existence of cultural differences in the effects of stress on the adolescent development. Cultures differ in the norms, beliefs, and values that provide prescriptions for behavior and therefore have a different impact on the stress process. Aldwin (2007) stated that culture can affect the stress and coping process in four ways. It shapes the types of stressors that a person is likely to experience, affects the appraisal of the stressfulness of a given situation, affects the choice of coping strategies, and provides different institutional mechanisms by which a person can cope with stress. Furthermore, because of the intercultural exchange that has increased markedly over the past several decades throughout the world, the problem of acculturative stress has become highly pronounced. Also, it has been emphasized that stress increases with rapid social change, most often in the form of westernization and a change in values (e.g., individualism, competition, and consumerism) (Dasen, 2000).

Considering ethnic differences in the stress consequences, Dornbusch et al. (1991) have found significant relationships between stressful life events and symptoms of depression among Asian and African American, Latino, and white adolescents, but the relationship was the weakest for African Americans. Similarly, Gerard and Buehler (2004) have found that the relationship between risk factors and depression in adolescence was moderated by racial–ethnic group membership and that this relationship was stronger for whites compared with both blacks and Latinos. Also, rates of completed as well as attempted suicide vary according to ethnicity. For example, the elevated rates of suicide among Native American youth are supposed to be associated with low social integration that results in limited access to various resources. On the other hand, lower rates of suicide among African American youth have been associated with their higher levels of religiosity compared with white youth. Highest levels of suicidal ideation and attempts are found in Hispanic females and can be attributed to a combination of cultural factors, including familism, authoritarian parenting style, and conflicts with caregivers over the issue of individuality (Jacobson and Gould, 2009). Comparing adolescent coping behaviors from seven different European nations, Gelhaar et al. (2007) found the highest cultural differences for coping with job-related problems, while similarities were found for coping with self- and future-related problems. Cross-cultural differences in quality and availability of social support also should be taken into account. Namely, despite social change, stress of adolescents is reduced more efficiently in those societies that manage to maintain a strong cultural identity and a part of their value system, such as family solidarity.

Conclusion

Although studies on the effects of stress on development of adolescence are numerous, more longitudinal and integrative research is needed to broaden our understanding of how complex relations between stress and mediating and moderating processes (personality, cognitive appraisal, coping, social support) exert their effects on (un)healthy development of adolescents. Also, the problem of the lack of comprehensive research exploring the factors associated with ethnicity that influence the relationship between stress and adolescent health is becoming more and more pronounced because the current intercultural exchange has considerably increased worldwide.

It is well known that stressful experience can lead to long-term changes in the body by destabilizing different physiological pathways, which often result in various mental and physical health outcomes. Therefore, knowledge of the deleterious effects of stress on adolescent health outcomes may be of help in planning and organizing prevention and treatment programs and thus improve adolescents' quality of life.

See also: Adolescence, Sociology of; Adolescent Health and Health Behaviors; Adolescent Sexual Risk; Alcohol Use among Young People; Alcohol Use and Abuse; Antisocial Behavior; Anxiety and Anxiety Disorders; Coping across the Lifespan; Depression, Pessimism, and Health; Depression; Suicide, Sociology of; Suicide.

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